Highlights

Machine Learning-Based Optimization of Football Squad Composition Under Budget Constraints

Lucio Vargas de Albuquerque Nunes, Dilson Lucas Pereira

- Machine learning predicts team performance from player statistics ($R^2 \approx 0.55$)
- Greedy and local search algorithms optimize squad composition under budget constraints
- Framework simulates realistic football transfer scenarios
- Experiments conducted using Premier League and Serie A datasets

Machine Learning-Based Optimization of Football Squad Composition Under Budget Constraints⋆

Lucio Vargas de Albuquerque Nunes[a], Dilson Lucas Pereira[a,*]

[a] *Universidade Federal de Lavras, Lavras, MG, Brazil*

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ABSTRACT

Football is characterized by high levels of uncertainty, which makes squad planning and player recruitment complex decision-making processes. This study investigates the use of Machine Learning (ML) and Operations Research (OR) techniques as decision-support tools for football squad composition. A predictive model based on a multilayer perceptron (MLP) is trained to estimate the final number of points obtained by football teams in a season using individual player statistics from the previous season. The predicted team performance is then used as the objective function in heuristic optimization algorithms that simulate player recruitment under budget constraints. Experiments conducted using data from major European leagues demonstrate the potential of combining predictive modeling and optimization to support squad planning decisions. The proposed framework provides a structured and data-driven approach to evaluating recruitment strategies, reducing subjectivity in player acquisition and contributing to more informed decision-making in professional football.

1. Introduction

Football is the most popular sport in the world Almulla, Islam, Al-Absi and Alam (2023). Over the past decades, the sport has undergone significant technological and analytical development, accompanied by substantial economic growth in the global football industry Rodrigues and Pinto (2023). Despite these advances, football remains highly unpredictable. A notable example is Leicester City's unexpected triumph in the Premier League (PL) in 2016, an outcome that surprised analysts and fans worldwide Baboota and Kaur (2019).

Given the economic relevance and global popularity of football, both academic researchers and industry professionals have increasingly explored data-driven approaches to support decision-making in the sport Baboota and Kaur (2019). The financial magnitude of football is reflected in the transfer market, which moves billions of dollars annually. For instance, between June and September 2023, global player transfers reached approximately USD 7.3 billion according to FIFA reports GLOBO Esporte (2023). These large financial investments highlight the importance of improving decision-making processes related to player recruitment and squad construction.

In the field of Computer Science, several studies have investigated the use of statistical analysis and machine learning (ML) techniques to model patterns and generate predictions in complex environments. Joseph, Fenton and Neil (2006), for example, investigated the use of Bayesian networks and other ML techniques to predict match outcomes for Tottenham Hotspur during the 1995–1997 seasons. In professional football practice, data-driven strategies have also been adopted by clubs such as Brighton & Hove Albion, which has implemented a recruitment model strongly supported by statistical analysis. This approach has enabled the identification of undervalued players in the transfer market, generating substantial financial returns through successful transfers BBC Sport (2023); CNN Brasil (2023); Olley (2023).

In academic research, most studies applying ML to football focus primarily on predicting match outcomes, estimating win probabilities, or optimizing betting strategies. However, comparatively fewer studies address the problem of squad composition and player recruitment using data-driven approaches Abidin (2021). To the best of our knowledge, few studies attempt to predict season-level team performance from individual player statistics and incorporate these predictions into a formal squad optimization framework. This gap is particularly relevant because

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* Corresponding author
luciovanunes@gmail.com (L.V.d.A. Nunes); dilson.pereira@ufla.br (D.L. Pereira)
https://orcid.org/0000-0001-8803-3293 (L.V.d.A. Nunes); https://orcid.org/0000-0002-6307-5152 (D.L. Pereira)
ORCID(s):

squad construction requires evaluating multiple technical, physical and contextual factors simultaneously, including team playing style, league characteristics and player career stage Abidin (2021). Furthermore, the stochastic nature of football performance means that even seemingly optimal transfers may fail to deliver the expected results.

Transfer market inefficiencies further illustrate the complexity of these decisions. According to Graham (2024), a transfer can be considered successful if the player starts at least 50% of matches during the first two seasons with the club, a concept referred to as the "50% rule". Based on this criterion, approximately 46% of transfers above EUR 10 million in the PL between 1992 and 2021 can be classified as unsuccessful Graham (2024). These outcomes are often associated with factors such as tactical incompatibility, inaccurate evaluation of player ability, positional mismatches, coaching decisions or physical issues.

Recent advances in ML have significantly expanded the capacity to analyze sports data, enabling the identification of patterns that were previously difficult to detect in large datasets. However, predictive models do not eliminate the inherent uncertainty of football; rather, they serve as decision-support tools that can help reduce risk and support strategic choices.

To address this challenge, this study proposes an integrated framework that combines machine learning-based performance prediction with heuristic squad optimization.

The main contributions of this work are threefold. First, we develop a machine learning model capable of predicting season-level team performance from individual player statistics. Second, we integrate these predictions into an optimization framework that simulates squad recruitment decisions under budget constraints. Third, we demonstrate through computational experiments that the proposed approach can identify recruitment strategies capable of improving predicted team performance.

2. Related Work

The application of ML in sports analytics has grown substantially over the past decades, particularly in football. According to Horvat and Job (2020), ML techniques are well suited for sports data analysis due to the large volume of structured and unstructured information generated from match events, tracking systems and historical databases. Consequently, numerous studies have investigated predictive models for football performance.

Most of the existing literature focuses on predicting the outcome of individual matches, typically framed as a classification problem involving win, draw or loss. Early work by Joseph, Fenton and Neil (2006) employed Bayesian networks to model match outcomes using a dataset of Tottenham Hotspur matches between 1995 and 1997. More recent studies have explored a broader range of machine learning algorithms and larger datasets. For example, Baboota and Kaur (2019) compared multiple supervised learning methods, including Random Forest, Support Vector Machines and Gradient Boosting, to predict PL match outcomes. Similarly, Awadallah and Khandelwal (2020) evaluated deep learning models such as dense neural networks and Long Short-Term Memory (LSTM) networks using data from ten PL seasons. These studies demonstrate the potential of ML techniques for modeling football outcomes, but they remain primarily focused on predicting individual matches.

Beyond match-level prediction, some research has investigated the determinants of team performance over an entire season. Statistical analyses have shown that aggregated performance indicators such as goals scored, shots on target and possession are associated with final league standings Lago-Ballesteros and Lago-Peñas (2010). Economic studies have also examined the relationship between financial investment and team performance, indicating that variables such as wage expenditure can influence competitive outcomes Carmichael, McHale and Thomas (2011). While these studies contribute to understanding factors associated with success in football leagues, they generally rely on retrospective analyses and do not explicitly model team performance as a function of the individual composition of the squad.

In parallel with predictive modeling, operations research (OR) techniques have been applied to various decision-making problems in sports. These applications include scheduling competitions, optimizing tournament structures and analyzing tactical decisions Kendall, Parkes and Spoerer (2010). Many of these problems involve combinatorial optimization and high computational complexity, often requiring heuristic or metaheuristic approaches to obtain feasible solutions.

More directly related to team composition, several studies have investigated optimization-based approaches to player selection. Tavana, Azizi, Azizi and Behzadian (2013) proposed a fuzzy inference system for selecting players and determining tactical formations, while Qader, Ahmed, Deb and Jindal (2017) formulated player selection as a multicriteria decision-making problem based on performance indicators. Other works have modeled squad selection using mathematical programming approaches, including stochastic programming models for team composition Pantuso

(2017) and assignment-based formulations for player-position allocation Boon and Sierksma (2003). Although these studies advance the formal modeling of squad selection problems, their objective functions are typically based on predefined performance scores or expert-defined evaluations.

Recent developments in prescriptive analytics emphasize the integration of predictive models with optimization methods to support decision-making processes Bertsimas and Kallus (2019). In this paradigm, machine learning models are not only used to generate predictions but also incorporated directly into optimization frameworks that guide strategic decisions.

Despite these advances, the integration between predictive models of team performance and optimization-based squad composition remains relatively unexplored in the football analytics literature. To the best of our knowledge, few studies attempt to predict season-level team performance directly from individual player statistics and incorporate these predictions into a formal squad optimization framework. This study addresses this gap by proposing a framework that combines machine learning-based prediction of team performance with an optimization procedure designed to simulate player recruitment decisions under budget constraints.

3. Data and Feature Engineering

The dataset used in this study consists of player performance statistics collected from two major European football leagues: the English PL and the Italian Serie A. These competitions were selected due to their high level of competitiveness and the availability of detailed player performance data.

The data were obtained from the FBref football statistics platform through web scraping procedures implemented in Python. The dataset covers five seasons, from 2019–2020 to 2023–2024, including all teams that participated in the first division of each league during this period. The dataset contains 200 team-season observations and 4,000 player-season records.

Each observation corresponds to the statistical performance of a player during a specific season while representing a particular team. The dataset contains 47 player performance attributes describing both offensive and defensive contributions, including advanced football statistics such as goals, assists, shots, passes and defensive actions.

To construct the predictive framework, player statistics from season $t-1$ were associated with the final number of points obtained by each team in season t. This temporal alignment ensures that all explanatory variables precede the target variable, preventing the use of future information during model training.

The two leagues were used separately during the modeling process. Specifically, the ML model was trained using data from one league and evaluated using data from the other. This cross-league strategy allows the assessment of model generalization across distinct competitive environments while avoiding information leakage between training and test samples.

All data processing and experiments were implemented in Python using standard ML libraries.

4. Performance Prediction Model

To predict team performance, a ML model was developed using individual player statistics as explanatory variables. Since the original dataset contains 47 performance attributes per player, dimensionality reduction was applied not only to mitigate multicollinearity among player statistics but also to reduce the dimensionality of the learning problem and improve model generalization.

The number of principal components was determined using a scree plot analysis. As shown in Figure 1, the cumulative explained variance reaches approximately 97.66% with 20 components, after which additional components provide only marginal gains. Therefore, 20 principal components were selected.

After dimensionality reduction, each player is represented by a vector of 20 principal components. To construct a team-level representation, each squad was truncated to the 20 players with the highest playing time during the season. Consequently, each team is represented by a vector containing the PCA components of its 20 players, resulting in 400 player-related features.

Additionally, the number of points obtained by the team in the previous season was included as an explanatory variable, producing a final input vector with 401 attributes.

The overall prediction framework used in this study is illustrated in Figure 2.

Because the ordering of players within a squad does not carry intrinsic meaning, a data augmentation strategy was employed to generate multiple equivalent representations of each team. This procedure consisted of randomly

Table 1
Performance comparison of machine learning models on the test dataset.

Model	R^2	MAE	MSE	RMSE	MAPE	Time (s)
LR	0.41	11.48	191.78	13.85	0.2580	1.72
RF	0.46	10.52	174.98	13.23	0.2390	118.66
XGBoost	0.41	11.23	191.23	13.83	0.2512	99.08
MLP	**0.55**	**11.08**	**170.42**	**13.05**	0.2424	161.68

permuting the ordering of players within the input vector, generating multiple training samples representing the same squad while preserving the original statistical properties of the dataset.

Several machine learning algorithms were evaluated for the task of predicting team performance from player statistics. In addition to the Multilayer Perceptron (MLP), experiments were conducted using Linear Regression (LR), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). These models represent different families of predictive approaches, including linear models, ensemble methods based on decision trees, and neural networks.

The comparative evaluation indicated that the MLP achieved the best overall predictive performance in terms of the coefficient of determination (R^2) and prediction error metrics. For this reason, the MLP was selected as the predictive component of the proposed framework and was used as the objective function in the subsequent squad optimization stage.

The selected MLP architecture consists of two hidden layers with 128 and 32 neurons, respectively, both using ReLU activation functions. Batch normalization and dropout layers were incorporated to improve generalization and mitigate overfitting. The output layer contains a single neuron with linear activation, producing the predicted number of points obtained by the team at the end of the season.

The models were trained using data from one league and evaluated using data from the other league, ensuring that training and testing were performed on distinct competitions. This strategy prevents information leakage between datasets and provides a more realistic evaluation of the model's generalization capability.

5. Predictive Performance

This section presents the performance of the machine learning models used to predict the final number of points obtained by football teams during a season based on individual player statistics from the previous season.

Several machine learning algorithms were evaluated for this prediction task, including LR, RF, XGBoost and MLP neural network. These models represent different families of predictive approaches, including linear models, tree-based ensemble methods, and neural networks.

Table 1 presents the predictive performance obtained by each model on the test dataset.

Among the evaluated models, the MLP achieved the highest coefficient of determination ($R^2 = 0.55$) and the lowest prediction errors in terms of MSE and RMSE. These results indicate that the neural network was able to capture nonlinear relationships between player statistics and team performance more effectively than the other tested models.

Based on this comparative evaluation, the MLP was selected as the predictive component of the proposed framework and used as the objective function in the subsequent squad optimization stage.

In regression problems, a natural baseline model consists of predicting the mean value of the target variable for all observations. In the context of this study, such a baseline corresponds to predicting the average number of points obtained by teams in the league for every sample.

Under this baseline, the coefficient of determination is $R^2 = 0$, meaning that the model does not explain any additional variance beyond the mean prediction. Therefore, positive values of R^2 indicate that the model is able to capture meaningful patterns in the data and outperform the baseline strategy.

Although the predictive model explains approximately 55% of the variance in team performance, the prediction task remains inherently challenging. Football outcomes are influenced by multiple contextual factors that are not directly captured by individual player statistics, such as tactical strategies, injuries, managerial decisions, and interactions between players during matches.

Nevertheless, the obtained results demonstrate that aggregated player performance indicators contain meaningful information about team-level outcomes, making the predictive model suitable for use as an objective function in the squad optimization framework proposed in this study.

5.1. MLP Training Behavior

Before analyzing the predictions generated by the model, the training dynamics of the neural network were examined. Figure 3 presents the evolution of the Mean Absolute Error (MAE) across training epochs for both the training and validation sets.

The MAE decreases rapidly during the initial training epochs, indicating that the model quickly learns relevant patterns from the data. After this initial phase, the error stabilizes with only small oscillations.

The proximity between the training and validation curves throughout the training process suggests that the model does not suffer from significant overfitting, indicating good generalization capacity.

5.2. Prediction Analysis

To better understand the predictions produced by the MLP model, the relationship between the real and predicted team points was analyzed.

Figure 4 shows the comparison between real and predicted points for teams in the PL across the evaluated seasons.

The results indicate that the model tends to produce more accurate predictions for teams positioned near the middle of the league table. This behavior is common in predictive models applied to highly uncertain environments: when the available variables do not capture the full variability of the phenomenon, predictions naturally converge toward intermediate values.

Consequently, the model tends to underestimate top-performing teams and overestimate low-performing teams, a phenomenon consistent with the statistical effect known as regression toward the mean.

To illustrate this behavior in more detail, Figures 5a and 5b present the comparison between real and predicted points for two clubs with contrasting performance levels.

The model successfully captures general performance trends, such as the improvement observed in Arsenal in recent seasons and the performance decline experienced by Southampton. However, the predictions exhibit reduced amplitude, with extreme performances being slightly underestimated or overestimated.

Based on these results, the MLP model was selected as the predictive function used in the squad optimization stage presented in the following section. Because the predictive model acts as a black-box objective function, gradient-based or exact optimization methods are not directly applicable. Therefore, heuristic search strategies were adopted to explore the solution space.

6. Squad Optimization

Once the ML model capable of predicting the final number of points obtained by a team in a season is identified, the second stage of the framework focuses on optimizing squad composition. In this phase, the problem is formulated as a constrained optimization task in which different recruitment strategies are evaluated according to the predicted team performance generated by the machine learning model.

As illustrated in Figure 6, the optimization stage begins with the current squad and a pool of candidate players. The algorithm simulates potential recruitment decisions and evaluates each resulting squad configuration using the machine learning model under budget constraints.

Let P denote the set of candidate players available in the transfer market and let S represent the current squad of a team. Each player $i \in P$ has an associated synthetic market value c_i and performance attributes derived from the previous season. The objective is to select a subset of players $X \subseteq P$ whose recruitment maximizes the predicted team performance while respecting the available transfer budget B.

Formally, the optimization problem can be expressed as:

$$\max_{X \subseteq P} f(S \cup X) \tag{1}$$

subject to:

$$\sum_{i \in X} c_i \leq B \tag{2}$$

where $f(\cdot)$ represents the machine learning model responsible for predicting the final number of points obtained by the team in the target season.

6.1. Synthetic Player Market Value

To incorporate financial constraints into the optimization process, a synthetic market value was defined for each player based on age and aggregated statistical performance. The value assigned to player i is computed as:

$$V_i = 1000 \cdot \left(1000 \cdot \max(0, 30 - I_i) + 10 \cdot \sum_{j=1}^{n} x_{ij} \right) \tag{3}$$

where I_i represents the age of player i and x_{ij} corresponds to the j-th statistical attribute associated with the player.

The age component assigns higher values to younger players, reflecting the typical valuation dynamics of football transfer markets, in which younger players tend to retain higher resale potential and longer career horizons. The second term aggregates player performance through the sum of available statistical indicators.

The external scaling factor is applied solely to produce values with magnitudes compatible with transfer budget constraints used during the optimization procedure. This synthetic valuation does not aim to reproduce real market prices, but rather to provide a consistent cost function suitable for the optimization framework proposed in this study.

6.2. Greedy Recruitment Algorithms

Two greedy heuristics were implemented to iteratively construct a set of player recruitments.

Starting from the current squad, the algorithm evaluates candidate players from the previous season while excluding athletes already present in the team or previously selected during the optimization process. Each candidate is evaluated using the machine learning model that predicts the final number of points obtained by the team.

To maintain a constant squad size during evaluation, newly recruited players replace the same number of players with the lowest playing time in the original squad. A candidate is considered feasible only if its synthetic market value respects the available transfer budget.

Two variants of the greedy algorithm were tested. In the *first-improvement* strategy, candidates are evaluated sequentially and the first player that improves the objective function is accepted. In the *best-improvement* strategy, all feasible candidates are evaluated and the player producing the largest improvement in predicted team performance is selected.

6.3. Local Search Optimization

In addition to greedy heuristics, local search algorithms were implemented to explore the solution space more thoroughly.

The procedure starts from a randomly generated feasible solution composed of players from the previous season whose combined market value respects the available budget. To ensure computational tractability, a pool of 100 candidate players is randomly sampled from the full dataset. This sampling strategy was adopted to maintain computational tractability while preserving diversity among candidate players.

The neighborhood of a solution is defined through swap operations, in which a player currently selected for recruitment is replaced by another candidate not yet chosen. A swap is considered feasible if the resulting squad respects the budget constraint.

Two acceptance strategies were evaluated. In the *first-improvement* variant, the search stops as soon as a better neighboring solution is found. In the *best-improvement* variant, all neighbors are evaluated and the swap producing the largest improvement in the objective function is selected.

The iterative search process terminates when no feasible swap produces further improvement in the predicted team performance, indicating convergence to a local optimum in the explored solution space.

Table 2
Preliminary benchmark results of the optimization algorithms.

Algorithm	Time (s)	Iterations	Evaluations	Improvements	Final score
Greedy First-Improvement	1.85	4	10	–	75.48
Greedy Best-Improvement	488.74	4	2493	–	**78.04**
Local Search First-Improvement	228.98	9	1076	8	77.24
Local Search Best-Improvement	114.34	3	584	2	77.66

Table 3
Summary of results across 10 independent runs.

Algorithm	Time (s)	σ_T	Iterations	Evaluations	Score	σ_S
Greedy First-Improvement	1.823	0.240	4.0	10.0	75.480	0.000
Greedy Best-Improvement	482.422	32.540	4.0	2493.0	**78.038**	0.000
Local Search First-Improvement	157.191	24.710	5.8	828.1	77.191	0.282
Local Search Best-Improvement	121.671	14.732	3.5	649.0	77.051	0.419

7. Squad Optimization Results

This section presents the experimental results obtained from the squad optimization procedures described in the previous section. The objective of the optimization stage is to identify player recruitment combinations that maximize the predicted team performance under budget constraints. The quality of each solution is evaluated using the predicted number of points returned by the machine learning model.

Since the optimized squads represent hypothetical recruitment scenarios, no real-world outcomes are available for comparison. Therefore, solution quality is measured exclusively through the objective function defined by the predictive model.

7.1. Preliminary Benchmark

An initial benchmark experiment was conducted to illustrate the behavior of the optimization algorithms. The experiment considered the Napoli squad in the 2021–2022 season with an available recruitment budget of EUR 120 million. The baseline predicted score of the original squad, estimated by the machine learning model, was 71.83 points.

Table 2 presents the results obtained after a single execution of each algorithm.

All algorithms improved the predicted team performance compared to the baseline score of 71.83 points. The Greedy Best-Improvement algorithm produced the highest predicted score (78.04 points), although at a significantly higher computational cost due to the large number of objective function evaluations.

The Greedy First-Improvement algorithm achieved the lowest execution time but produced solutions of lower quality. Local search algorithms obtained intermediate results in terms of predicted score, with Local Search Best-Improvement requiring fewer evaluations than the First-Improvement variant.

7.2. Multiple Runs Analysis

To obtain a more robust comparison between the algorithms, ten independent executions were performed for each method. Table 3 summarizes the results, including average execution time, number of iterations, objective function evaluations and predicted score.

The greedy algorithms exhibit a clear trade-off between solution quality and computational cost. Greedy Best-Improvement achieved the highest average score (78.038 points), but required substantially more evaluations than Greedy First-Improvement.

In contrast, the local search algorithms presented lower computational cost while achieving competitive solution quality. Local Search Best-Improvement required fewer evaluations than the First-Improvement variant and showed similar predicted scores.

The greedy algorithms produced deterministic results under the experimental configuration, resulting in zero standard deviation across runs. The local search methods showed small variability due to the random initialization of the search process.

Table 4
Optimization results for Cagliari with a budget of EUR 60 million.

Algorithm	Time (s)	Iterations	Evaluations	Improvements	Final score
Greedy First-Improvement	3.93	5	21	–	42.74
Greedy Best-Improvement	553.14	5	2702	–	43.04
Local Search First-Improvement	90.23	12	424	11	**43.69**
Local Search Best-Improvement	76.16	5	363	4	43.36

7.3. Budget Sensitivity Analysis

To analyze the impact of financial constraints on the optimization process, an additional experiment was conducted using a smaller budget. The experiment considered the Cagliari squad with an available budget of EUR 60 million. The baseline predicted score for the original squad was 42.56 points.

Table 4 presents the results obtained under this restricted scenario.

All algorithms improved the predicted team performance compared to the baseline. In this more restrictive scenario, the local search methods produced the best solutions, with Local Search First-Improvement achieving the highest predicted score (43.69 points).

These results suggest that when the budget constraint is tighter, the iterative exploration performed by local search algorithms may become more effective than purely constructive greedy strategies.

Overall, the experiments demonstrate that combining machine learning predictions with heuristic optimization can identify recruitment strategies capable of improving the expected performance of football teams under realistic financial constraints.

8. Discussion

The results obtained in this study demonstrate the potential of combining machine learning and optimization techniques to support decision-making in football squad construction. The predictive model was able to explain approximately 55% of the variability in the final league points of teams based solely on individual player performance statistics from the previous season. Considering the inherent uncertainty of football competitions and the absence of tactical and contextual variables in the model, this level of explanatory power suggests that aggregated player statistics contain meaningful information about collective team performance.

The optimization experiments illustrate how these predictions can be incorporated into decision-support tools for player recruitment. The results show that heuristic algorithms are capable of identifying player combinations that improve the predicted performance of teams while respecting financial constraints. In scenarios with larger budgets, greedy algorithms were able to identify high-quality solutions, whereas under tighter financial constraints local search methods produced more competitive results.

These findings highlight the importance of considering both predictive accuracy and search strategy when integrating machine learning models into optimization frameworks. The predictive model defines the quality landscape explored by the algorithms, while the optimization procedure determines how efficiently promising solutions are discovered within the search space.

Overall, the proposed framework demonstrates how performance prediction models can be combined with heuristic optimization to simulate player recruitment scenarios and support data-driven decision-making in professional football.

However, the proposed framework still presents some limitations. The predictive model relies exclusively on individual player statistics and does not explicitly capture tactical interactions between players or contextual factors such as injuries and managerial decisions. Future work may explore richer representations of team dynamics to improve predictive accuracy.

9. Conclusion

This study proposed a framework that integrates machine learning and optimization techniques to support football squad planning. A predictive model based on a multilayer perceptron was developed to estimate team performance using individual player statistics from the previous season. The model achieved an explanatory power of approximately $R^2 = 0.55$, indicating its ability to capture relevant patterns linking player performance and collective team outcomes.

The predicted team performance was then used as the objective function in a squad optimization problem under budget constraints. Experimental results showed that heuristic algorithms can identify recruitment strategies capable of improving the predicted performance of teams while respecting financial limitations.

These results suggest that integrating predictive modeling and optimization can provide a structured and data-driven approach to squad construction in professional football. By transforming player performance data into quantitative decision-support tools, the proposed framework contributes to reducing subjectivity in recruitment processes and advancing the application of analytical methods in sports management.

Future work may explore the incorporation of additional contextual variables, such as tactical information and player interactions, as well as the application of more advanced optimization techniques to further improve squad selection strategies.

Declaration of Generative AI Use

The authors declare that generative artificial intelligence tools were used to improve the clarity, fluency, and organization of the manuscript. These tools were employed exclusively for language refinement and structural assistance, without contributing to the scientific content, methodology, or results of the study. All intellectual contributions, analyses, and conclusions presented in this work are solely those of the authors, who take full responsibility for the content of the manuscript.

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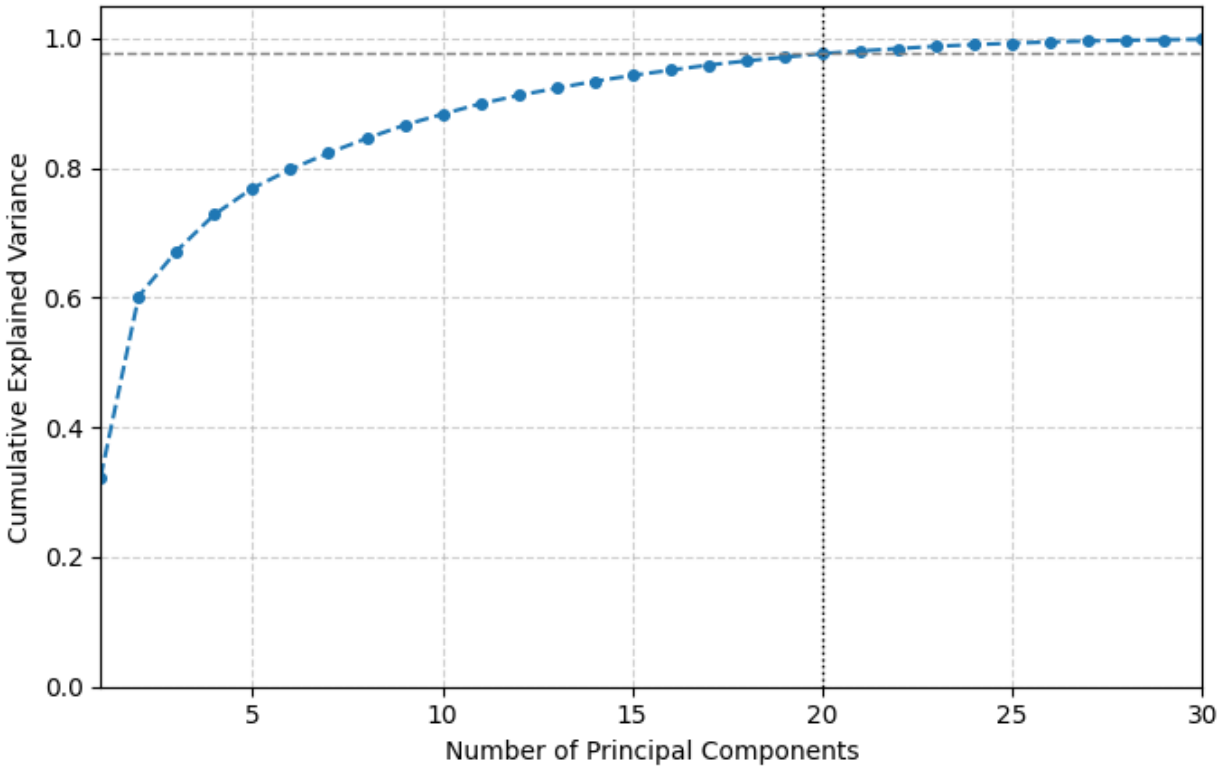


Figure 1: Cumulative explained variance obtained through principal component analysis (PCA). The first 20 components explain approximately 97.66% of the total variance.

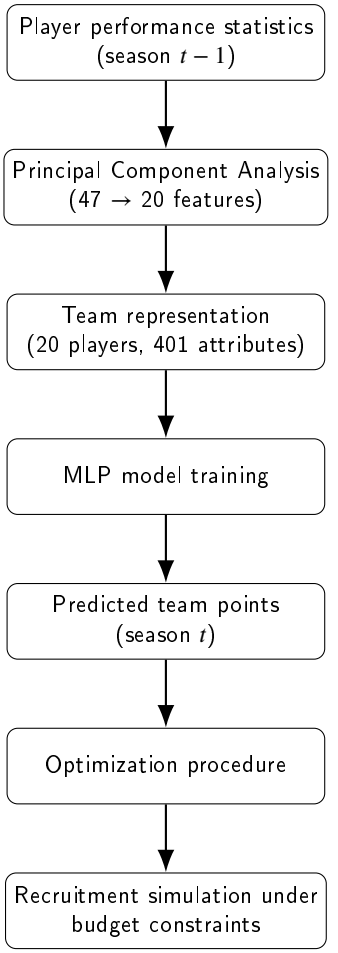


Figure 2: Overview of the proposed framework integrating machine learning-based performance prediction and squad optimization.

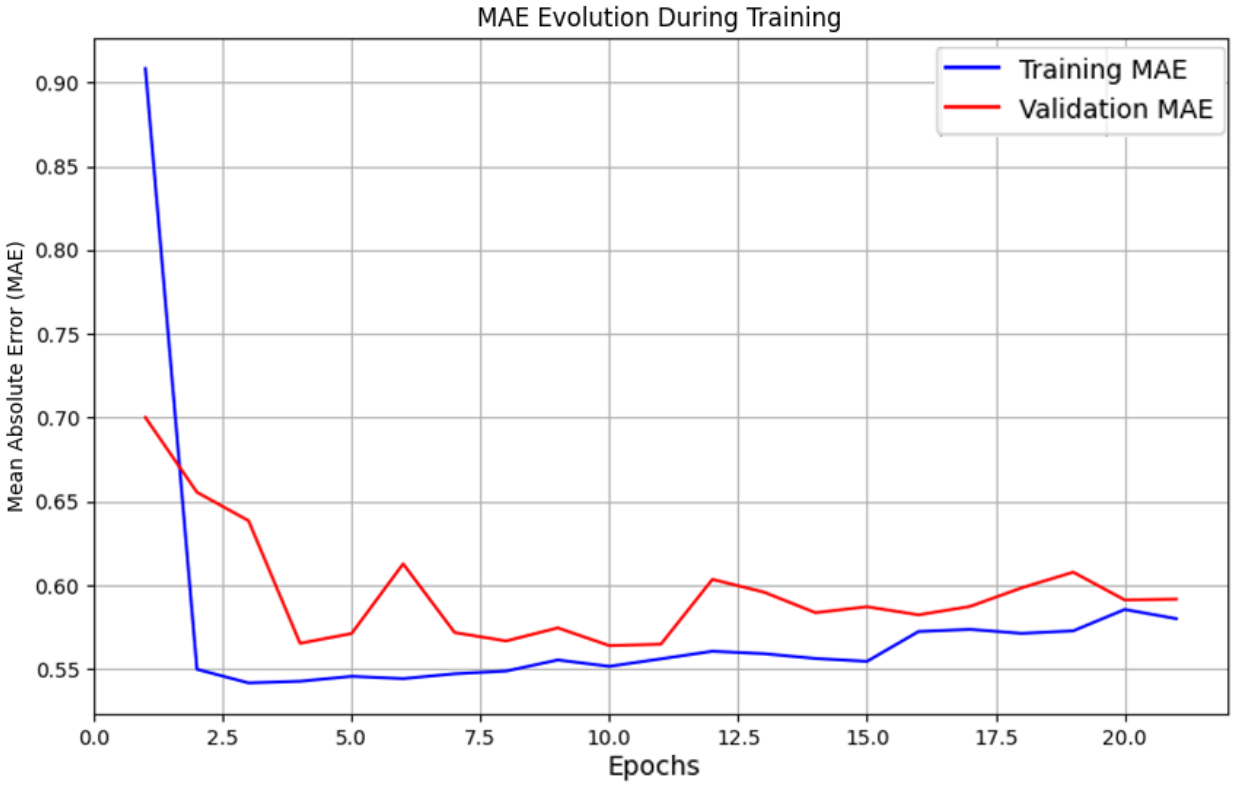


Figure 3: Evolution of the mean absolute error (MAE) during the training of the MLP model.

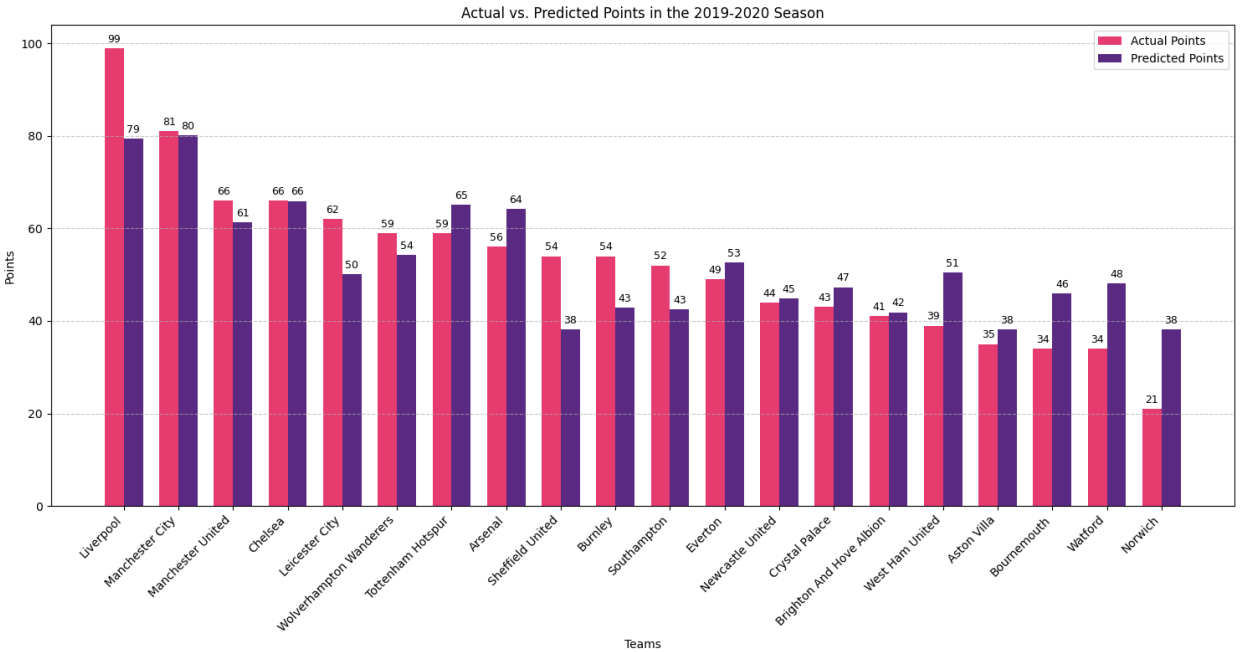
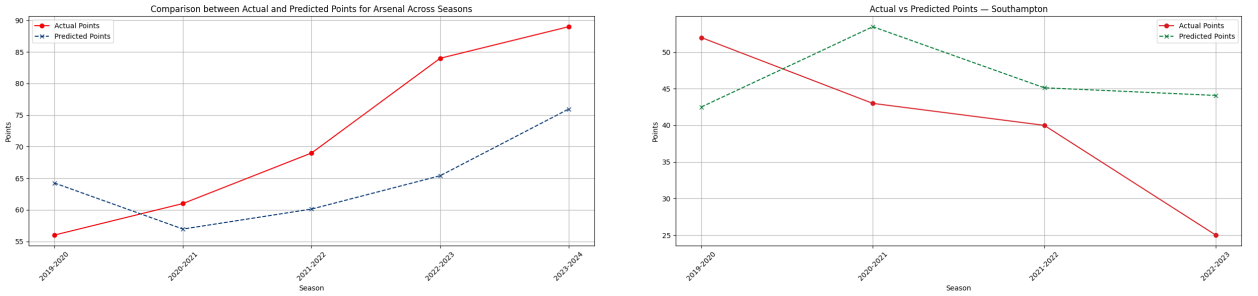


Figure 4: Comparison between actual and predicted team points in the Premier League.



(a) Arsenal (b) Southampton

Figure 5: Comparison between actual and predicted team points across seasons for selected clubs.

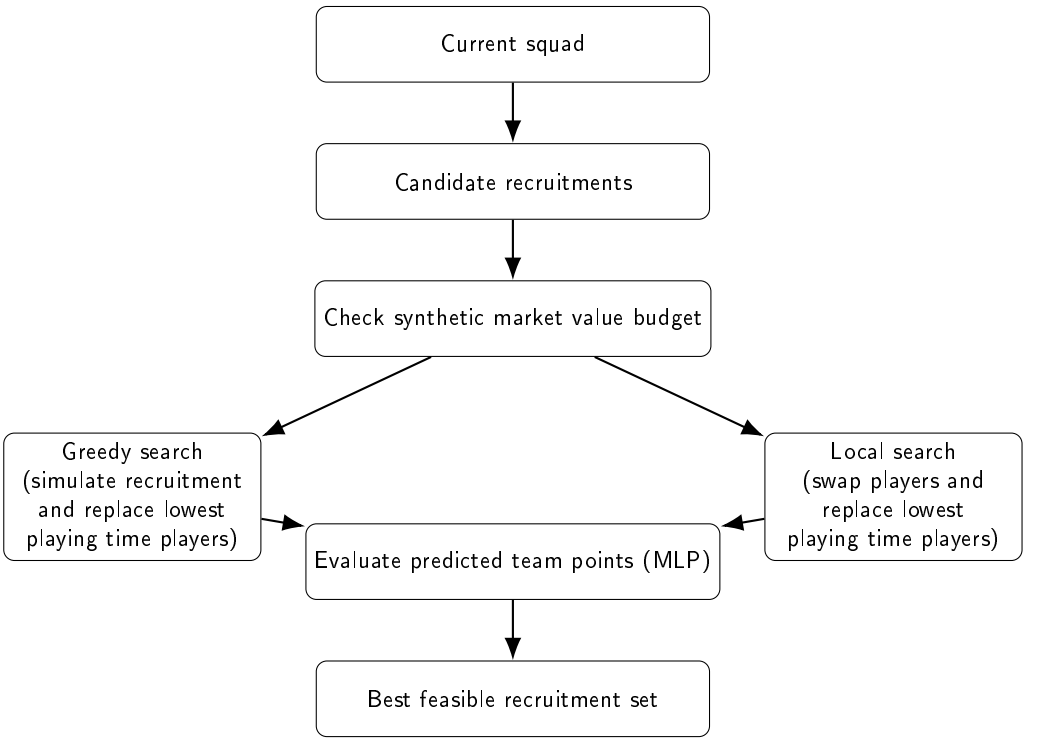


Figure 6: Pipeline of the squad optimization procedure. Recruitment simulations are evaluated using the machine learning model under budget constraints.